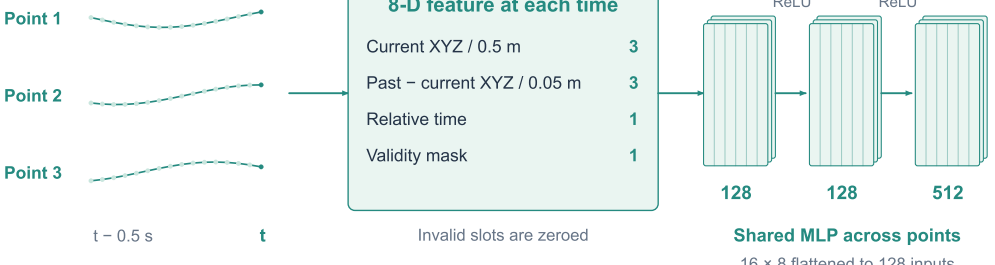
Trajectory encoding and experimental controls

Implemented design: fixed causal windows, shared ACT core and an auxiliary-gradient control

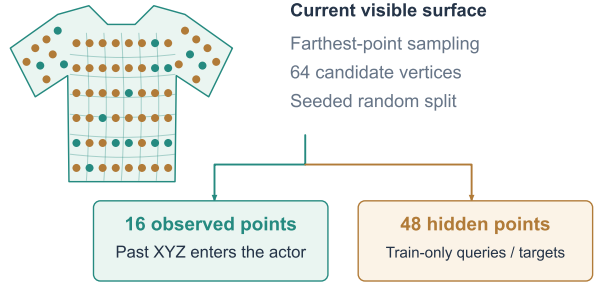
a Point-wise trajectory representation

Persistent point identities within each window



K = 16 trajectories, T = 16 slots

b Observed / hidden point split



Resample identities across windows, retain identity across time

c Matched observation controls

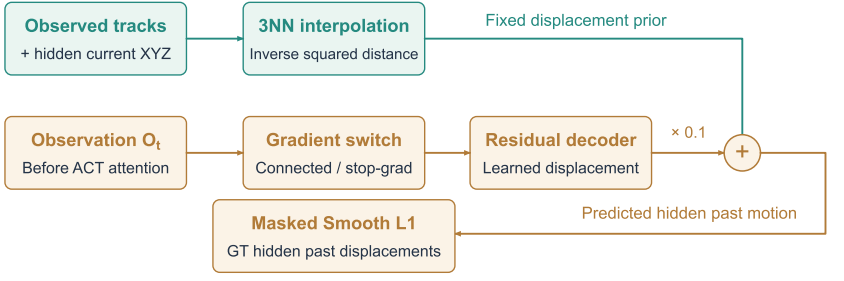
Every condition uses current RGB and current joint state

Condition	RGB history	Current XYZ	Past motion	Aux. loss
ACT	—	—	—	—
RGB-H	16 frames	—	—	—
Geometry	16 frames	✓	—	—
Tracks	16 frames	✓	✓	—
Tracks + Completion	16 frames	✓	✓	✓

T = 16 at stride 3, 90 Hz. Predict 100 joint targets in every condition.
History-enabled variants share the RGB window. GT point tracks are privileged inputs.

d 3NN residual completion control

Same actor inputs and architecture in both variants



Connected: auxiliary gradients reach shared observation encoders.
Detached: the auxiliary head learns, with its memory input detached.